

SHUNYA OS

FDA1–FDA36 — FINAL DIRECTIVES

Master execution playbook • Founder finality • Git / CI / staging / production governance built in

Purpose: this is the final directive library for completing SHUNYA OS as a production-grade, business-capable platform. The FDA naming replaces the previous Directive 01–36 naming. FDA means *Final Directive Architecture*: these are execution directives, not suggestions or a feature wishlist.

Finality principle: the project is not complete because code exists or tests are green. It is complete only when the promised user outcome works end-to-end, survives failure, is secure, deployable, recoverable, observable, auditable and usable on supported devices.

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Operating system for execution

Every FDA inherits BOTH contracts below.

A. Engineering / product contract

- Inspect the existing repository, runtime and canonical architecture before changing code.
- Reuse canonical systems; do not create a parallel production identity, event, memory, execution or audit system.
- Build real production behavior: no mock, fake, placeholder or disconnected implementation presented as complete.
- Integrate backend, persistence, permissions, UI, APIs, observability and failure handling as one capability.
- Prefer deterministic computation and free/open/self-hosted components; isolate unavoidable paid/vendor dependencies.
- Protect tenant isolation, secrets, privacy, authorization and AI/tool boundaries.
- Add unit, integration, end-to-end and negative/failure tests appropriate to the capability.
- Commit meaningful work and push it. Never commit OAuth tokens, credentials, runtime data or secrets.
- Report exact files, tests, runtime evidence, Git commit, push state and unresolved findings.
- Do not declare completion if a promised user outcome still depends on manual intervention merely because the code path exists.

B. Git / deployment / release contract

- **Git checkpoint:** at directive start record branch, HEAD, origin relation and working-tree state; never overwrite unrelated founder work.
- **Commit discipline:** commit each coherent completed work unit; every directive must end with a meaningful commit unless explicitly blocked.
- **Push discipline:** push every completed directive to the canonical remote; verify remote HEAD matches the intended commit.
- **Secret discipline:** credentials, OAuth tokens, runtime stores, local databases, backups and secrets remain ignored and are explicitly checked before commit.
- **Staging deployment:** after each directive, deploy the pushed commit to the designated staging/verification environment when infrastructure supports it; run smoke tests there.
- **Production deployment:** do not automatically expose unfinished directive work publicly. Production deploy follows batch certification, or an explicitly approved hotfix path.
- **Migration safety:** before schema/data migrations or destructive changes, take/verify a restorable backup and document rollback.
- **Deployment evidence:** record commit, build/version, environment, migration status, health checks and smoke-test result.
- **Rollback readiness:** every deployment-affecting directive must identify how to revert code and data safely.
- **No dirty-tree handoff:** Hermes must not claim completion while uncommitted material exists without explicitly identifying it and why it is intentionally left untouched.

Required cadence

Moment	Mandatory action
Before FDA	Inspect repository, architecture, branch/HEAD, origin, dirty tree, active services, backups and current deployment.
During FDA	Work in coherent commits. Test continuously. Never mix unrelated cleanup with capability work.
End of each workstream	Run relevant unit/integration/negative tests; record failures rather than hiding them.
End of FDA	Full relevant tests → Git diff/check → secret check → commit → push → verify remote → staging deploy → staging smoke test → evidence
After FDA 05/10/15/20/25/30/35	Mandatory forensic review. Remediation before next batch.
Production	Batch-certified changes only, except explicitly governed hotfixes. Backup/migration/health/rollback evidence required.
After production deploy	Verify release commit, health, core smoke path, logs/metrics, error rate and rollback readiness.
Before final launch	Fresh-environment deployment rehearsal, full simulation, backup restore, security/performance/accessibility checks and founder acceptance

FDA1 — Canonical Architecture & Ownership Lock

Objective: Make the repository's architectural truth explicit before expanding capability.

Mandatory workstreams

1. Inventory every active module and map each core concept to exactly one canonical owner.
2. Define ownership for identity, tenant, objects, events, observations, evidence, memory, commitments, BusinessExecutionInstance, tasks, plans, decisions, execution, learning, conversations, documents, integrations and audit.
3. Identify legacy, duplicate, shadow and archived paths; label which are production, transitional or archive-only.
4. Produce dependency and data-flow maps from ingestion through decision and execution.
5. Define rules for adding future modules without creating parallel authorities.

Acceptance gates

- No critical concept has ambiguous production ownership.
- Archived code cannot be imported accidentally into production.
- Canonical architecture document is machine-checkable or otherwise enforceable.
- Regression tests prove canonical imports/routes remain stable.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

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FDA2 — Core Runtime Consolidation

Objective: Create one authoritative event → observation → awareness → decision → execution → evidence → learning path.

Mandatory workstreams

1. Trace all existing runtime paths and remove or quarantine duplicate production pipelines.
2. Unify event IDs, observation IDs, execution IDs and evidence references.
3. Ensure BusinessExecutionInstance remains the durable execution aggregate where applicable.
4. Make retries, idempotency, failure and partial execution explicit.
5. Verify legacy routes cannot silently become alternate production paths.

Acceptance gates

- One E2E path can be traced from input to outcome and audit evidence.
- Duplicate runtime behavior is removed or explicitly archived.
- Failure and retry semantics are tested.

FDA closure package

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- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA3 — Canonical Memory & Knowledge

Objective: Turn SHUNYA memory into one durable, governed, searchable company context layer.

Mandatory workstreams

1. Define structured memory, episodic/event memory, relationship memory, commitment memory and document knowledge boundaries.
2. Store provenance, source, timestamps, confidence, tenant, retention and correction history.
3. Support retrieval, update, correction, deletion and export.
4. Prevent transient JSON/files from becoming accidental canonical memory.
5. Define what AI may remember automatically versus what requires user/system confirmation.

Acceptance gates

- Memory retrieval is deterministic enough to test.
- Corrections propagate safely.
- Tenant isolation and deletion semantics are proven.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA4 — Identity & Relationship Integrity

Objective: Make people, companies, leads, customers, suppliers and other entities resolve consistently.

Mandatory workstreams

1. Centralize identity resolution and alias handling.
2. Implement deduplication, merge, split and conflict workflows.
3. Attach source provenance and confidence to identity claims.
4. Ensure Gmail/contacts/import/API sources all converge on the same identity system.
5. Test historical data when identities change.

Acceptance gates

- Duplicate entities can be safely merged and reversed where required.
- Conflicting sources are visible rather than silently overwritten.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA5 — API & Integration Fabric

Objective: Create a durable integration foundation for every later connector.

Mandatory workstreams

1. Define versioned REST/OpenAPI conventions and error contracts.
2. Build OAuth/token lifecycle abstractions, scopes and credential health.
3. Implement webhook verification, idempotency, replay protection and retries.
4. Create connector health/status diagnostics.
5. Define provider-neutral internal contracts so vendors can be replaced.
6. Add import/export APIs and audit hooks.

Acceptance gates

- A connector can fail without corrupting core state.
- API documentation and contract tests exist.
- Secrets never enter source control.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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MANDATORY BATCH GATE — AFTER FDA5

Freeze forward development. Perform Forensic Review #1. Check architecture, functionality, data integrity, AI behavior, integrations, UX, security, performance, testing, Git history, deployment evidence, backup/rollback and founder outcome. Issue one consolidated remediation directive if required. Only then continue.

FDA6 — SHUNYA Intelligence Core

Objective: Deliver evidence-backed reasoning and decision support across the canonical data model.

Mandatory workstreams

1. Create context assembly using company data first.
2. Support classification, summarization, comparison, recommendation, prioritization and explanation.
3. Expose evidence, assumptions and confidence.
4. Separate deterministic facts from inference.
5. Respect authorization and tenant context during retrieval.

Acceptance gates

- Identical test inputs produce stable deterministic portions.
- Recommendations cite supporting records.
- Permission boundaries are tested.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA7 — Web Intelligence

Objective: Add reliable external research without allowing web content to become authority.

Mandatory workstreams

1. Implement search/retrieval adapters with source provenance.
2. Track freshness and source timestamps.
3. Handle conflicting sources and low-quality results.
4. Provide citations/links for external claims.
5. Isolate fetched content from tool permissions and system instructions.
6. Cache where appropriate without presenting stale information as current.

Acceptance gates

- A user can see why a web claim was made.
- Prompt injection in web content cannot authorize an action.
- Provider failure degrades gracefully.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA8 — Model Orchestration

Objective: Make SHUNYA resilient across free/open/local and optional external models.

Mandatory workstreams

1. Create model abstraction and capability registry.
2. Route deterministic work away from LLMs.
3. Add free/open/local model routes where quality is sufficient.
4. Add controlled fallback and escalation.
5. Implement spend, latency and availability policies.
6. Log model/version/cost metadata without exposing secrets.

Acceptance gates

- No single model provider is a hard business-critical dependency where avoidable.
- Fallback behavior is tested.
- Paid calls are explicit and policy-controlled.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
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- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA9 — Founder Intelligence Cockpit

Objective: Make SHUNYA answer the founder's highest-value questions from evidence.

Mandatory workstreams

1. Build 'what changed', 'what needs me', 'what is at risk', 'where is money going', 'which customers need attention', 'what opportunities exist' views.
2. Prioritize rather than dump data.
3. Link every important conclusion to underlying records.
4. Support drill-down from executive signal to source evidence.
5. Keep the interface calm and contextual.

Acceptance gates

- Founder can reach the next action without hunting across modules.
- Signals have source/evidence links.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA10 — Proactive Awareness

Objective: Surface useful changes without notification overload.

Mandatory workstreams

1. Define alert classes, severity, deduplication and suppression.
2. Monitor deadlines, commitments, customer risk, cash risk, operational exceptions and important external changes.
3. Create actionable alerts with owner, reason and next action.
4. Support snooze, acknowledge, resolve and audit.
5. Prevent repeated or low-value notifications.

Acceptance gates

- Alerts are explainable and actionable.
- No alert storm under repeated events.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
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MANDATORY BATCH GATE — AFTER FDA10

Freeze forward development. Perform Forensic Review #2. Check architecture, functionality, data integrity, AI behavior, integrations, UX, security, performance, testing, Git history, deployment evidence, backup/rollback and founder outcome. Issue one consolidated remediation directive if required. Only then continue.

FDA11 — CRM Foundation

Objective: Complete lead-to-customer relationship management.

Mandatory workstreams

1. Capture leads from supported channels.
2. Resolve identity and ownership.
3. Implement qualification, SLA, rotation/reassignment, follow-up and pipeline.
4. Support opportunity/proposal/quote stages and loss reasons.
5. Maintain complete relationship history.

Acceptance gates

- A real lead can progress to customer without a disconnected manual bridge.
- SLA and ownership behavior are tested.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA12 — Sales Intelligence

Objective: Make sales execution intelligent and measurable.

Mandatory workstreams

1. Lead scoring and qualification.
2. Next-best action and follow-up suggestions.
3. Pipeline health and forecast.
4. Salesperson workload and SLA analytics.
5. Conversion/loss analysis.
6. Evidence-backed recommendations rather than opaque scores.

Acceptance gates

- Sales decisions can be explained.
- Forecast inputs are traceable.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA13 — Customer Experience

Objective: Make the customer relationship continuous rather than transaction-based.

Mandatory workstreams

1. Customer profile, preferences and history.
2. Onboarding and service commitments.
3. Communications and issue tracking.
4. Satisfaction/complaint/escalation handling.
5. Retention, repeat and referral intelligence.
6. Customer-visible status where appropriate.

Acceptance gates

- Customer history is available from one context.
- Escalations become governed commitments/tasks.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
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FDA14 — Marketing OS

Objective: Create the operational marketing system.

Mandatory workstreams

1. Campaigns, audiences, channels and content planning.
2. Lead capture and campaign metadata.
3. UTM/source tracking.
4. Landing/lead capture integration points.
5. Approvals and publishing states.
6. Asset/document linkage.

Acceptance gates

- Campaign can be created, executed, measured and linked to leads.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA15 — Marketing Intelligence

Objective: Connect marketing activity to revenue outcomes.

Mandatory workstreams

1. Build campaign → lead → customer → revenue attribution.
2. Track conversion, CAC/CPA where meaningful, referral and repeat contribution.
3. Compare channels and experiments.
4. Surface wasted spend and high-performing segments.
5. Keep attribution assumptions explicit.

Acceptance gates

- Revenue attribution can be audited back to source events.

FDA closure package

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- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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- Working-tree state and every intentionally uncommitted item.
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MANDATORY BATCH GATE — AFTER FDA15

Freeze forward development. Perform Forensic Review #3. Check architecture, functionality, data integrity, AI behavior, integrations, UX, security, performance, testing, Git history, deployment evidence, backup/rollback and founder outcome. Issue one consolidated remediation directive if required. Only then continue.

FDA16 — Operations OS

Objective: Turn commitments into coordinated execution.

Mandatory workstreams

1. Tasks, plans, projects, dependencies and SLAs.
2. BusinessExecutionInstance integration.
3. Owner/approver assignment.
4. Exception handling and escalation.
5. Supplier/customer dependencies.
6. Execution evidence and completion criteria.

Acceptance gates

- A commitment can be followed from creation to verified outcome.
- Partial completion is represented honestly.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA17 — Procurement & Supplier Intelligence

Objective: Control vendor-side execution and cost.

Mandatory workstreams

1. Supplier master data and identity.
2. Quote comparison and selection evidence.
3. Purchase requests and approvals.
4. Purchase commitments and delivery tracking.
5. Vendor performance, pricing history and risk.
6. Supplier communication linkage.

Acceptance gates

- Procurement decision is auditable from request to outcome.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA18 — Finance Core

Objective: Build real operational finance workflows.

Mandatory workstreams

1. Invoices, bills, expenses, receivables and payables.
2. Approval and payment status.
3. Customer/supplier financial linkage.
4. Document/evidence linkage.
5. Profitability and financial period concepts.
6. Accounting-provider adapter boundary.

Acceptance gates

- Financial state is persistent and reconciliable.
- No financial action bypasses required authorization.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA19 — Cash & Reconciliation

Objective: Make cash truth operational.

Mandatory workstreams

1. Payment capture/status integration.
2. Bank/feed/import abstraction where available.
3. Reconciliation workflows.
4. Unmatched transaction handling.
5. Cash position and forecast.
6. Duplicate payment protection and audit.

Acceptance gates

- Reconciliation exceptions are visible and actionable.
- Payment status is not inferred solely from a UI click.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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FDA20 — Tax & Compliance

Objective: Prepare compliance reliably without pretending to replace licensed professionals.

Mandatory workstreams

1. Versioned tax/compliance rules.
2. Compliance calendar and deadlines.
3. Evidence collection and reconciliation.
4. GST/TDS or jurisdiction-specific workflows where supported.
5. Filing preparation/export.
6. Professional review/approval boundary.

Acceptance gates

- Rule versions are traceable.
- Compliance outputs show source/rule version and review state.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
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MANDATORY BATCH GATE — AFTER FDA20

Freeze forward development. Perform Forensic Review #4. Check architecture, functionality, data integrity, AI behavior, integrations, UX, security, performance, testing, Git history, deployment evidence, backup/rollback and founder outcome. Issue one consolidated remediation directive if required. Only then continue.

FDA21 — Audit & Governance

Objective: Make consequential activity provable.

Mandatory workstreams

1. Immutable or append-only audit events where appropriate.
2. Decision traces and execution traces.
3. Approval history.
4. Evidence chains.
5. Audit exports and professional review packages.
6. Access/change logs.

Acceptance gates

- A material business outcome can be reconstructed after the fact.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
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FDA22 — Admin & Permissions

Objective: Complete multi-tenant administration and least privilege.

Mandatory workstreams

1. Tenants, organizations and business units.
2. Users, roles, permissions and scopes.
3. Approval matrices and delegation.
4. Service accounts and connector permissions.
5. Configuration and policy management.
6. Tenant isolation tests.

Acceptance gates

- Unauthorized cross-tenant access is demonstrably blocked.
- Permission changes are audited.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

Never declare the FDA complete from Hermes' summary alone. The closure evidence must be independently reviewable.

FDA23 — People / Internal Operations

Objective: Support required internal workforce workflows.

Mandatory workstreams

1. Roles/responsibilities, workload and internal tasks.
2. Attendance/leave where business-required.
3. Policy/SOP acknowledgement.
4. Training records.
5. Internal approvals and escalation.
6. Privacy-aware access boundaries.

Acceptance gates

- Only necessary people data is collected and exposed.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

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FDA24 — Document & Knowledge OS

Objective: Make documents usable, searchable and governed.

Mandatory workstreams

1. Ingestion, metadata, classification and OCR where needed.
2. Versioning and permissions.
3. Search/retrieval and citations.
4. Retention/deletion.
5. Relationship to entities, commitments and evidence.
6. Prompt-injection isolation for untrusted documents.

Acceptance gates

- A document can become evidence/context without becoming an unauthorized instruction.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

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FDA25 — Universal Import / Export / Migration

Objective: Allow businesses to bring history in and take governed data out.

Mandatory workstreams

1. CSV/XLSX/JSON/document ingestion.
2. Preview, schema mapping and validation.
3. Identity resolution and deduplication.
4. Dry-run and rollback.
5. Progress/error reporting.
6. Export with permissions and provenance.

Acceptance gates

- A bad import cannot silently corrupt canonical data.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

Never declare the FDA complete from Hermes' summary alone. The closure evidence must be independently reviewable.

MANDATORY BATCH GATE — AFTER FDA25

Freeze forward development. Perform Forensic Review #5. Check architecture, functionality, data integrity, AI behavior, integrations, UX, security, performance, testing, Git history, deployment evidence, backup/rollback and founder outcome. Issue one consolidated remediation directive if required. Only then continue.

FDA26 — Developer / Integration Platform

Objective: Make SHUNYA extensible without architectural drift.

Mandatory workstreams

1. OpenAPI documentation.
2. Webhook/event contracts.
3. Connector SDK conventions.
4. Versioning/deprecation policy.
5. Developer diagnostics.
6. Sandbox/test fixtures that are clearly non-production.
7. Integration health dashboard.

Acceptance gates

- A new connector can be added without creating a second identity/event architecture.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

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FDA27 — Web App Completion

Objective: Finish the actual public and authenticated product.

Mandatory workstreams

1. Public shunyaos.com experience.
2. Authenticated app.shunyaos.com.
3. Authentication, onboarding, workspace and settings.
4. Core object/workspace/conversation surfaces.
5. Search, notifications and admin.
6. Loading, empty, error and recovery states.
7. No dead routes or placeholder screens.

Acceptance gates

- Every visible action either works or is deliberately absent from release.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

Never declare the FDA complete from Hermes' summary alone. The closure evidence must be independently reviewable.

FDA28 — Responsive & Accessibility Completion

Objective: Make the product device-independent and accessible.

Mandatory workstreams

1. Desktop, laptop, tablet and mobile layouts.
2. Portrait mobile behavior.
3. Keyboard navigation.
4. Screen-reader semantics.
5. Focus management.
6. Contrast and readable typography.
7. Touch target and overflow testing.
8. Browser matrix certification.

Acceptance gates

- No core workflow breaks at supported dimensions.
- Accessibility defects are launch-gated, not cosmetic backlog items.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

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FDA29 — Observability & Support

Objective: Make production behavior visible and supportable.

Mandatory workstreams

1. Structured logs with correlation IDs.
2. Metrics for requests, jobs, queues, integrations and failures.
3. Tracing across important workflows.
4. Health/readiness endpoints.
5. Alerting and incident signals.
6. Operator diagnostics and runbooks.
7. User-facing status/recovery where appropriate.

Acceptance gates

- A production incident can be diagnosed without SSH archaeology.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

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FDA30 — Security & AI Safety

Objective: Establish production security and safe AI execution.

Mandatory workstreams

1. Threat model and trust boundaries.
2. Secret management and rotation.
3. Tenant isolation.
4. AuthN/AuthZ and session security.
5. Input/output validation.
6. Prompt-injection and data-exfiltration defenses.
7. Tool permission gating.
8. Rate limits and abuse controls.
9. Dependency/security scanning.
10. Privacy and retention controls.

Acceptance gates

- No unresolved critical/high launch blocker.
- Untrusted content cannot elevate privileges.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

Never declare the FDA complete from Hermes' summary alone. The closure evidence must be independently reviewable.

MANDATORY BATCH GATE — AFTER FDA30

Freeze forward development. Perform Forensic Review #6. Check architecture, functionality, data integrity, AI behavior, integrations, UX, security, performance, testing, Git history, deployment evidence, backup/rollback and founder outcome. Issue one consolidated remediation directive if required. Only then continue.

FDA31 — Backup & Disaster Recovery

Objective: Prove SHUNYA can recover from serious failure.

Mandatory workstreams

1. Automated backups with retention policy.
2. Encrypted backup storage.
3. Restore into clean environment.
4. Database and object-store consistency.
5. Recovery runbook.
6. RPO/RTO targets.
7. Periodic restore test.

Acceptance gates

- A real restore has been successfully demonstrated, not merely configured.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

Never declare the FDA complete from Hermes' summary alone. The closure evidence must be independently reviewable.

FDA32 — Performance & Scale

Objective: Make performance predictable before public growth.

Mandatory workstreams

1. Define latency budgets for key workflows.
2. Load test APIs, queues and ingestion.
3. Test large imports/documents.
4. Queue backpressure and worker scaling.
5. Database query/index review.
6. Caching strategy.
7. Graceful degradation under provider/API pressure.

Acceptance gates

- Performance budgets are measured and documented.
- No known single-node bottleneck blocks expected initial scale.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

Never declare the FDA complete from Hermes' summary alone. The closure evidence must be independently reviewable.

FDA33 — Deployment & Release Engineering

Objective: Make releases reproducible and reversible.

Mandatory workstreams

1. Environment separation.
2. Dependency lock/verification.
3. Database migration discipline.
4. Build/deploy scripts.
5. Health checks and smoke tests.
6. Rollback procedure.
7. Release notes and versioning.
8. CI checks and protected release path.

Acceptance gates

- A clean deployment can be reproduced from documented steps.
- Rollback is tested.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

Never declare the FDA complete from Hermes' summary alone. The closure evidence must be independently reviewable.

FDA34 — Full Business Simulation

Objective: Test SHUNYA as a company, not as isolated modules.

Mandatory workstreams

1. Run marketing → lead → sales → customer → operations → supplier → invoice → payment → finance → tax → audit → founder cockpit.
2. Inject duplicates, missing data, conflicting identity and late commitments.
3. Inject provider outage, expired OAuth, webhook duplication and partial execution.
4. Verify evidence, notifications, permissions and recovery.
5. Run representative executive queries at the end.

Acceptance gates

- The complete lifecycle works through real production paths.
- Injected failures recover without corrupting truth.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

Never declare the FDA complete from Hermes' summary alone. The closure evidence must be independently reviewable.

FDA35 — Public Launch Rehearsal

Objective: Rehearse the public launch in a clean production-like environment.

Mandatory workstreams

1. Fresh environment from release artifacts.
2. Seed/configuration procedure.
3. Domain/TLS/authentication checks.
4. OAuth consent and connector setup.
5. New-tenant onboarding.
6. Core business simulation.
7. Monitoring/alerts.
8. Backup/restore check.
9. Support/incident drill.
10. Rollback drill.
11. Founder walkthrough on desktop and mobile.

Acceptance gates

- No launch blocker remains hidden behind an already-configured development machine.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

Never declare the FDA complete from Hermes' summary alone. The closure evidence must be independently reviewable.

MANDATORY BATCH GATE — AFTER FDA35

Freeze forward development. Perform Forensic Review #7. Check architecture, functionality, data integrity, AI behavior, integrations, UX, security, performance, testing, Git history, deployment evidence, backup/rollback and founder outcome. Issue one consolidated remediation directive if required. Only then continue.

FDA36 — Final Certification & Maintenance Lock

Objective: Certify SHUNYA and transition from construction to maintenance/growth.

Mandatory workstreams

1. Freeze feature scope.
2. Run full regression and E2E suite.
3. Review security, performance, accessibility and integration evidence.
4. Verify Git release commit, remote state and clean release tree.
5. Verify backups and restore evidence.
6. Verify documentation and operator runbooks.
7. Founder acceptance in clean environment.
8. Record known non-blocking maintenance items separately from missing capability.
9. Tag release and establish maintenance/change-control policy.

Acceptance gates

- No known foundational capability required by the public promise is incomplete.
- Public launch can proceed with confidence.
- Post-launch work is maintenance, provider/regulatory adaptation or growth-led improvement — not unfinished core construction.

FDA closure package

- Exact files/modules changed and canonical owners touched.
- Tests run, including negative/failure tests where relevant.
- Runtime/UI/API evidence for the promised user outcome.
- Security/privacy/tenant-impact review.
- Git: starting HEAD → final commit; diff/check results; commit hash; push result; remote HEAD verification.
- Deployment: environment/version; migration/backup state if applicable; health and smoke-test result; rollback path.
- Working-tree state and every intentionally uncommitted item.
- Known limitations, with each classified as launch blocker, maintenance, provider dependency or explicitly out of scope.

Never declare the FDA complete from Hermes' summary alone. The closure evidence must be independently reviewable.

Forensic review checklist

Area	Independent verification
Architecture	No duplicate production owner, runtime, memory, identity or execution path.
Functionality	Real user workflow works end-to-end; no mock or disconnected screen.
Data	Persistence, provenance, deduplication, correction and tenant isolation are correct.
AI	Facts vs inference; sources; confidence; model routing; tool permissions; prompt-injection resistance.
Integrations	OAuth/API/webhook behavior, expiry, retries, duplication and provider outages.
UX	Desktop/mobile, loading/empty/error/recovery/accessibility states.
Security	Secrets, auth, authorization, isolation, injection, data exfiltration and abuse controls.
Reliability	Idempotency, retries, queues, partial failure and recovery.
Performance	Measured budgets and load behavior.
Git	Commit/push history is coherent; no unexplained dirty work.
Deployment	Staging/production version matches intended commit; migrations and rollback are known.
Recovery	Backup exists and restore path is proven where required.
Founder result	Capability reduces work/risk rather than moving it into another hidden manual process.

Batch certification rule: a failing review is not a reason to proceed. It becomes remediation work.

Final launch gate — FDA36

FDA36 closes the construction program only when all of the following are evidenced:

- Canonical release commit/tag exists and is pushed to the remote.
- Release tree is clean, with generated/local artifacts explicitly excluded or governed.
- Fresh-environment deployment succeeds from documented release artifacts.
- Database/data migrations are reproducible and rollback-safe.
- Backups and a real restore test are evidenced.
- Full regression, integration, E2E, failure-injection, security, performance and accessibility checks pass their defined gates.
- Public and authenticated web surfaces work across supported desktop/mobile dimensions.
- Critical free/open integrations work; provider failures degrade safely.
- Full business simulation works from marketing → lead → sales → customer → operations → supplier → invoice → payment → finance → tax → audit → founder.
- Support/incident/runbooks exist and have been exercised.
- Founder walkthrough succeeds in a clean environment.
- No promised foundational capability remains incomplete.

After FDA36: the default mode changes from construction to maintenance and growth. Future work is handled as versioned change requests, regulatory/provider updates, security fixes, reliability improvements or growth experiments — not as hidden unfinished core functionality.

One-time finality test: if the founder can launch publicly, operate the business, recover from failure, audit consequential activity, and focus on growth without discovering a missing foundational capability, SHUNYA has reached the intended completion state.